AWS NETWORKING PROJECT

VPC Peering

Connecting two AWS VPCs and testing private communication between EC2 instances

PROJECT REPORT

Submitted by
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AWS VPC • EC2 • Route Tables • Security Groups • SSH • ICMP

AWS Region: US East (N. Virginia) — us-east-1
Project architecture and instance details are based on the lab information shared during this conversation.

VPC Peering Architecture Overview

Supplied project visual



1. Project Objective

The objective of this project is to establish private communication between two separate Amazon Virtual Private Clouds (VPCs) using an AWS VPC Peering connection. The project uses a public subnet in VPC-A for a bastion host and private subnets for the application instances. App-A and App-B communicate using private IPv4 addresses rather than public internet addresses.

The project demonstrates CIDR planning, subnet placement, route-table configuration on both sides of a peering connection, security-group rules, SSH access, and ICMP-based connectivity testing.

2. Architecture and IP Address Plan

VPC-A contains one public subnet and one private subnet. VPC-B contains one private subnet. The bastion host is placed in VPC-A's public subnet and is used as an administration entry point. App-A and App-B are private application instances in their respective VPCs.

Resource	VPC / Subnet	CIDR or private IP
VPC-A	VPC-A	10.10.0.0/16
Public subnet A	VPC-A	10.10.1.0/24
Bastion host	Public subnet A	10.10.1.93
Private subnet A	VPC-A	10.10.2.0/24
App-A	Private subnet A	10.10.2.153
VPC-B	VPC-B	10.20.0.0/16
Private subnet B	VPC-B	10.20.1.0/24
App-B	Private subnet B	10.20.1.191

Bastion public IPv4 observed: 98.81.25.69. The application instances are addressed by their private IPs in the tests described here.

3. AWS Services and Components

Component	Purpose in this project
Amazon VPC	Provides isolated virtual networks for VPC-A and VPC-B.
Subnets	Separate public administration access from private application workloads.
Internet Gateway	Provides internet connectivity for the public subnet in VPC-A; it is not used as the VPC peering path.
EC2 instances	Bastion, App-A and App-B provide the hosts used in the lab.
VPC Peering	Connects VPC-A and VPC-B for private IPv4 communication.
Route tables	Direct traffic for the remote VPC CIDR to the peering connection.
Security Groups	Control allowed inbound traffic, including SSH and ICMP ping.
SSH key pair	Authenticates SSH sessions to Linux EC2 instances.

4. Implementation Summary

Step 1 — Create VPC-A. Create VPC-A with CIDR 10.10.0.0/16. Create a public subnet 10.10.1.0/24 and a private subnet 10.10.2.0/24.

Step 2 — Configure public access for the bastion. Attach an Internet Gateway to VPC-A and configure the public subnet's route table with a default route (0.0.0.0/0) to that Internet Gateway. Launch the bastion host in the public subnet.

Step 3 — Create VPC-B. Create VPC-B with CIDR 10.20.0.0/16 and its private subnet 10.20.1.0/24. The design shared for this project does not include an Internet Gateway in VPC-B.

Step 4 — Launch the application instances. Launch App-A in VPC-A's private subnet and App-B in VPC-B's private subnet. The private IPs used in the lab were 10.10.2.153 and 10.20.1.191 respectively.

Step 5 — Create and accept VPC peering. Request a peering connection between VPC-A and VPC-B, then accept it. Confirm that its state is Active before configuring routes.

Step 6 — Configure routes on both sides. In the route table associated with App-A's subnet, add destination 10.20.0.0/16 with the peering connection as target. In the route table associated with App-B's subnet, add destination 10.10.0.0/16 with the same peering connection as target. These two routes provide forward and return paths.

Step 7 — Configure security groups. Allow SSH (TCP port 22) only from the required administration source. For ping tests, allow ICMP Echo Request from the peer instance's private IP (/32) in the destination instance's security group. Add the reverse Echo Request rule if testing ping in the opposite direction.

5. Security Design

Security Groups act as stateful virtual firewalls. Rules should follow the principle of least privilege by restricting sources to the required private IP addresses or trusted security groups where supported. Avoid opening SSH or ICMP to 0.0.0.0/0 for this lab.

Traffic / purpose	Protocol and port	Example source restriction
SSH to App-B from App-A	TCP 22	10.10.2.153/32
Ping App-B from App-A	ICMPv4 Echo Request	10.10.2.153/32
Ping App-A from App-B	ICMPv4 Echo Request	10.20.1.191/32

The private key file (file1.pem) was present on the bastion host and had restrictive read-only permissions for its owner (equivalent to chmod 400). The lab also showed why a key file must exist at the path specified in the SSH command: an SSH attempt from App-A failed when file1.pem was not present in App-A's home directory.

6. Connectivity Testing and Observations

SSH test. A shell prompt for App-B was observed after connecting to 10.20.1.191, and the App-B login message identified the previous login source as 10.10.2.153. This supports that private SSH connectivity from App-A to App-B was achieved during the lab.

Command used for the App-A to App-B SSH test:

```
ssh -i file1.pem ec2-user@10.20.1.191
```

Ping test — App-A to App-B:

```
ping -c 4 10.20.1.191
```

Ping test — App-B to App-A:

```
ping -c 4 10.10.2.153
```

Observed status: The ping commands were started in both directions, but the terminal output shared in the conversation did not show successful Echo Reply packets. Therefore, ICMP connectivity is recorded as **not yet verified**, rather than as a successful test. The next troubleshooting step is to check the destination security group's ICMPv4 Echo Request rule, both subnet route tables, and network ACLs.

7. Troubleshooting Notes

Symptom	Checks / explanation
SSH connection timed out	Check instance state, public/bastion access path, subnet route table, security group TCP 22, and network AC
Permission denied (publickey)	Confirm the correct key pair, username (ec2-user for Amazon Linux), and that the private key is available to t
Identity file not accessible	The .pem file is not in the current directory or the specified path is incorrect. Use the correct path or an approv
Ping receives no replies	Check ICMP Echo Request in the destination security group, routes in both VPCs, network ACLs, and correc
Peering is Active but traffic fails	Active means the peering link is established; it does not automatically add routes or permit traffic.

8. Key Concepts Learned

• VPC CIDR ranges must not overlap for VPC peering.
• A peering connection creates a private network link, but route tables on both sides are still required.
• The return route is as important as the forward route.
• Security Groups control which traffic is permitted; SSH uses TCP port 22, while ping uses ICMP.
• A private EC2 instance can communicate with another VPC over peering without needing a public IP address.

• VPC peering is not transitive and does not provide VPC-B with internet access through VPC-A.

9. Conclusion

This project provided practical experience with AWS VPC networking and private communication between two VPCs. The lab established the architecture using VPC-A, VPC-B, a bastion host, two private application instances, and a VPC peering connection. SSH access from App-A to App-B was observed. ICMP ping testing was attempted in both directions, but successful ping replies were not evidenced in the shared terminal output and remain a verification item.

10. Cleanup Recommendations

After saving project evidence, remove lab resources that are no longer needed to avoid ongoing charges. Terminate EC2 instances, delete the peering connection and peer routes, remove lab security groups and subnets, detach and delete the Internet Gateway, and then delete the VPCs. Check for retained EBS volumes and Elastic IP addresses before finishing.

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